Meta-Probing Utility Predictor (MPUP): Predicting LLM-Specific Retrieval Utility for RAG without Fine-Tuning

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Abstract—Retrieval-Augmented Generation (RAG) systems rank passages by topical relevance, yet relevance and usefulness are not the same thing. A passage that matches a query may still fail to help the model answer. Recent utility-aware RAG methods proven to be impractical because existing methods either fine-tune a reranker or retriever's embedding model for every target LLM, or probe internal logits, which takes up high cost, memory, and computation. Meta-Probing Utility Predictor (MPUP) is a method to predict LLM-specific retrieval utility without fine-tuning the target model.

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I. Introduction

Large language models (LLMs) are Transformer-based neural networks pre-trained on massive unlabeled text so that they are adapted to specific tasks through prompting rather than task-specific retraining [Min et al., 2023], making one model can do reasoning, summarization, and generation tasks that previously require separate architectures [Chang et al., 2024]. LLMs are used in multiple areas, one of them is LLM for personalized tutoring and automated feedback in education [Kasneci et al., 2023]. LLMs can also be used in reshaping professional practice, research workflows, and policy [Dwivedi et al., 2023].

LLMs often produce hallucinations which are outputs unfaithful to the source or factually incorrect [Ji et al., 2023], because their knowledge is stored and frozen at training time, making them cannot be corrected without retraining [Huang et al., 2025]. Retrieval-Augmented Generation (RAG) addresses this problem by retrieving relevant documents from external sources and do generation on that retrieved evidence, so the model draws on up-to-date information instead of only memorized parameters [Fan et al., 2024]. RAGs have multiple methods to retrieve the documents, one of them is traditional RAG which select documents purely by topical relevance where the contents match the queries, which works worse than RAG systems where the documents are selected by their utility for the model [Zhang et al., 2026].

Selecting documents or passages based on their utility in RAG is done by seeing whether the passage providing it as evidence measurably improves a LLM's ability to generate an accurate answer compared to answering without any evidence [Zhang, et al., 2026]. To measure this, two cases need to be compared. In the first, the model answers from its internal memory alone. In the second, the same model answers with a specific retrieved passage available to it. A standard correctness metric scores both answers, and a passage counts as useful only when the second answer is more accurate than the first. Applying the same test to a large pool of candidates leaves only the passages that actually help that model [Zhang et al., 2026].

However, how can a practitioner estimate passage utility under realistic, low-budget conditions? Existing methods are too expensive for practical deployment, because they require massive 32-billion-parameter models [Zhang et al., 2025b], completely retraining the retriever's embedding model using complex objectives [Sandhu et al., 2026], or running the target LLM twice for every single query

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[Zhang et al., 2025c].

There are cheap alternatives method compared to LLM sampling. Semantic Perplexity (SePer) measures how useful a retrieved passage by observing how much its semantic perplexity changes after the passage is provided, but it still requires running an evaluation model, which adds computation and latency, making it less suitable for real-time passage selection [Dai et al., 2025]. Another approach proposes a lightweight neural model that predicts passage utility for a target Question Answering (QA) model, but its predictor is trained for a specific QA model, making it difficult to apply to other QA models [Perez-Beltrachini and Lapata, 2025]. Another method is a feature-based predictor combining Query Performance Prediction (QPP), perplexity, readability, and document quality features using linear regression to predict retrieval utility, but this method achieves relatively low accuracy for utility prediction, with Spearman $\rho \approx 0.21$–0.24, compared with the higher performance of methods that directly involve the LLM in utility estimation with Spearmen $\rho \approx 0.60+$ [Tian et al., 2026]. These methods have been proven that they haven't solved the problem yet, shown by the low accuracy and unapplicable solutions.

Several other methods have been explored. There are LURE-RAG which introduces a lightweight LambdaMART reranker trained with a listwise utility loss to optimize context ordering, but relying on complex listwise utility supervision requires generating query and passage combinations, making the labeling cost expensive [LURE-RAG, 2026]. Another approach is RRPO, where passages selection treated as a sequential decision process trained via reinforcement learning directly from LLM feedback, which also which takes up big part of the memory and computation [RRPO, 2026]. The current existing methods either require substantial memory and computational resources, or incur high costs.

This paper closes the gap with Meta-Probing Utility Predictor (MPUP). MPUP is a lightweight method to predict LLM-specific retrieval utility which means it can select the most useful passages for models without requiring massive computing power methods including fine tunings. Instead of digging into model's internal states and do methods like logit probing [Dai et al., 2025], MPUP evaluates a passage's usefulness by looking at 14 practical clues, such as document statistics, meaning overlap, and factual accuracy. MPUP compresses all the processing into highly efficient 4-bit precision so that the system can run smoothly on a single consumer-grade GPU rather than the multi-GPU or datacenter-scale hardware that fine-tuning-based alternatives require.

This paper makes three contributions. The first is that many cheap features, used together, can be as accurate as one expensive feature that probes the target LLM directly. They also fit within the memory of a single consumer GPU. The second is a predictor that works for several target LLMs at once. Because it does not have to be retrained when the LLM changes, switching models no longer means rebuilding the predictor. The third is that this predictor can be trained on annotations that already exist in the dataset, which are cheaper than labels produced by running the target LLM over every passage [Zhang et al., 2025a].

The remainder of this paper is organized as follows. Section II reviews related work on retrieval-augmented generation, utility-based passage selection, and lightweight utility prediction. Section III describes the proposed Meta-Probing Utility Predictor, covering the 14 features, the predictor architecture, and the 4-bit quantized setup. Section IV presents the experimental setup, datasets, and evaluation metrics, followed by the results and analysis, while Section V concludes the paper and outlines directions for future work.

II. Literature Review

LLM can only use the knowledge it learned during training, and this knowledge cannot be updated without retraining. RAG solves this by first retrieving documents from an external source, then ask the LLM to write its answer based on those documents [Fan et al., 2024]. In traditional RAG, passages are chosen by relevance, which means how closely their content matches the query.

However, a relevant passage is not always a helpful one. Recent work shows that choosing passages by their utility for the model gives better results than choosing them by relevance alone [Zhang et al., 2026]. Utility is measured by comparing two answers from the same LLM. The first answer is written from the model's own memory. The second is written with the candidate passage. Correctness metric scores both answers, and the passage is useful only if the second answer is more accurate [Zhang et al., 2026].

Utility also depends on what the model already knows. Passage may help one LLM but add nothing for another LLM that already knows the answer. So, utility belongs to a pair of passage and model, not to the passage alone. To estimate it in practice, we can either test the target LLM directly or approximate it with cheaper signals.

Methods that estimate passage utility can be grouped into four types. The first is utility estimation by large models or utility-trained retrievers. [Zhang et al., 2025b] estimate passage utility with 32-billion-parameter models, and [Sandhu et al., 2026] retrain the retriever's embedding model with complex objectives so that its ranking follows utility. Both reach utility-aware selection, but at the cost of heavy training and large models, which is hard to afford under low-budget conditions. The second is learned rerankers and selection policies trained from utility feedback. LURE-RAG introduces a lightweight LambdaMART reranker trained with a listwise utility loss [LURE-RAG, 2026], while RRPO treats passage selection as a sequential decision process optimized by reinforcement learning from LLM feedback [RRPO, 2026]. The former depends on

expensive query-passage labeling, and the latter consumes substantial memory and computation.

The third is utility estimation by directly involving the target LLM. [Zhang et al., 2025c] run the target LLM twice for every query to compare its behavior with and without the passage, and SePer measures how much the semantic perplexity changes once the passage is provided [Dai et al., 2025]. These methods are closest to the definition of utility as improvement over the model's internal memory, but they add inference cost and latency at selection time, so they are less suitable for real-time use. The fourth is lightweight utility predictors. [Perez-Beltrachini and Lapata, 2025] propose a lightweight neural model that predicts utility for a target QA model, yet it is trained for one specific model and is difficult to transfer to others. [Tian et al., 2026] combine Query Performance Prediction (QPP), perplexity, readability, and document quality features in a linear regression, which is cheap but reaches only Spearman $\rho \approx 0.21$–0.24, far below the $\rho \approx 0.60+$ of methods that involve the LLM directly.

MPUP is compared to eight previous studies which were based on six criteria, presented in Table 2.1. C1 is a lightweight model that runs on limited hardware. C2 is no fine-tuning of the LLM or the retriever. C3 is no additional LLM run for every query at selection time. C4 is utility estimation accuracy that is competitive with methods that probe the target LLM directly. C5 is a single predictor that works for several target LLMs without retraining. C6 is training on annotations that already exist in the dataset instead of labels generated by the target LLM.

TABLE I
Positioning of MPUP to Previous Studies

Method	C1	C2	C3	C4	C5	C6
Zhang et al., 2025b	×	×	✓	×	×	✓
Sandhu et al., 2026	×	×	✓	×	×	✓
Zhang et al., 2025c	×	✓	×	✓	×	✓
Dai et al., 2025 (SePer)	×	✓	×	✓	×	✓
Perez-Beltrachini & Lapata, 2025	✓	✓	✓	×	×	✓
Tian et al., 2026	✓	✓	✓	×	×	×
LURE-RAG, 2026	✓	✓	✓	×	×	✓
RRPO, 2026	×	×	✓	×	×	✓
MPUP (ours)	✓	✓	✓	✓	✓	✓

Table 2.1 shows that no previous study satisfies all six criteria. Methods that reach high accuracy through the target LLM [Zhang et al., 2025c; Dai et al., 2025] fail the lightweight and per-query cost criteria, while lightweight methods [Perez-Beltrachini and Lapata, 2025; Tian et al., 2026; LURE-RAG, 2026] either remain tied to one model or lose accuracy. None of the eight studies provides a predictor that transfers across several target LLMs (C5).

Utility-based selection is only useful if people can actually run it. Many practitioners work with a single consumer-grade GPU and a limited budget. Methods that need very large models, retriever retraining, or two LLM runs for every query are out of reach for them. Methods that are cheap but tied to one LLM create another problem, because the predictor must be rebuilt whenever the LLM is replaced. Lightweight methods with low accuracy are also not helpful, since they often pick the wrong passages. Predictor that is accurate, lightweight, and reusable across LLMs removes these barriers. It lets practitioners select passages that truly help their model without heavy computation and without rebuilding the predictor for each new LLM. MPUP is designed for this purpose by combining many cheap features into one predictor.

III. Methodology

Fig. 3.1 shows the MPUP pipeline. For each query, MPUP first asks an LLM M to write one short hypothetical answer. Then, for every candidate passage, it extracts 14 features from five feature families. A trained regressor maps these features to a utility score, and an optional calibration step adjusts the scale of the score. The passages are ranked by the score, and the top passages are placed in the prompt of the target LLM. The target LLM is never fine-tuned, and it does not need to answer the question twice for every passage.

Fig. 1. Overview of MPUP. LLM steps with M (shaded) run in 4-bit precision; the target LLM is not fine-tuned.

Passage utility measures whether a passage, given as evidence, improves the accuracy of an LLM compared with answering without any evidence [Zhang et al., 2026]. Let q be query, d candidate passage, and L target LLM. The utility of d for L on query q is the difference in generation accuracy [Zhang et al., 2026]:

$$u(d; L, q) = \mathrm{Acc}(L(q, d)) - \mathrm{Acc}(L(q, \varnothing)). \tag{1}$$

Where $L(q, \varnothing)$ is the answer of L from its own memory alone, and $L(q, d)$ is the answer of L when it reads d. $\mathrm{Acc}(\cdot)$ is a standard correctness metric. Passage is useful only if reading it improves the model, which means $u > 0$. A value of zero means the passage adds nothing, and a negative value means it makes the answer worse.

If we test a pool of candidate passages $\mathcal{C}$ in this way and keep only the passages that help L, we obtain the gold utilitarian set of L [Zhang et al., 2026]:

$$\widetilde{\mathcal{G}}_q^{(L)} = \{d \in \mathcal{C} \mid u(d; L, q) > 0\}. \tag{2}$$

Acc is the average of relaxed exact match and token-level F1 against the gold answers:

$$\mathrm{Acc}(a) = 0.5 \cdot \mathrm{EM}_r(a) + 0.5 \cdot \mathrm{F1}(a). \tag{3}$$

Relaxed exact match counts an answer as correct when a gold answer appears inside it as a token span. We use it because instruction-tuned LLMs answer in full sentences, and strict exact match would mark most of them wrong,

which pushes almost every utility to zero. The value of u lies in $[-1, 1]$.

Building $\widetilde{\mathcal{G}}$ directly needs two LLM runs for every query-passage pair, which is too expensive for real-time selection. MPUP predicts the utility instead, so that passages in $\widetilde{\mathcal{G}}$ are ranked first:

$$\hat{u}(d; q) = f_\theta(\phi(q, d; M)), \tag{4}$$

where $\phi(q, d; M) \in \mathbb{R}^{14}$ is the feature vector and f_θ is the predictor. M is the LLM used for the two LLM-based steps, the hypothetical answer and logit probing. It can be the target LLM L or a fixed model. In our experiments, M is Qwen2.5-7B-Instruct. The predictor f_θ is trained once, and it is not retrained when L changes.

The 14 features are listed in Table 3.1. D1 describes how the retriever scored the passage: its BM25 score (divided by the highest score of the query), its dense similarity, its rank, and its score gap to the next passage. D2 describes the passage itself: token length, Flesch–Kincaid grade, and a source credibility score. D3 measures meaning overlap. Entity overlap is the fraction of query entities, found by spaCy, that also appear in the passage. Answerability is the fraction of content words of the hypothetical answer that appear in the passage. D4 checks faithfulness with a DeBERTa-v3-small NLI model. The passage is the premise and the hypothetical answer is the hypothesis, and we keep the entailment and contradiction probabilities. D5 probes the LLM M. We compute the entropy of its next-token distribution at the end of two prompts: the query alone (H_{base}) and the passage followed by the query (H_{ctx}). The features are the two perplexities $\exp(H)$ and the difference $\Delta H = H_{\text{base}} - H_{\text{ctx}}$. The hypothetical answer is generated once per query and shared by D3 and D4.

TABLE II
Features of MPUP

Group	Features
D1 Retriever	BM25 score, dense similarity, rank, score gap
D2 Document quality	Token length, Flesch–Kincaid grade, source credibility
D3 Semantic overlap	Entity overlap, HyDE answerability
D4 Faithfulness	Entailment probability, contradiction probability
D5 LLM probing	Base perplexity, context perplexity, ΔH

The features are standardized and passed to a gradient-boosted regression tree model (200 trees, depth 5, learning rate 0.05, subsample 0.8). The features are few, tabular, and on different scales, so a tree ensemble is a simple and cheap choice.

The predictor is trained on labels derived from existing annotations. RAGBench marks which sentences of each document were used to produce the answer. The label of passage d is the fraction of its sentences that are marked:

$$\tilde{u}(d) = \frac{|S_d \cap S_{\text{used}}|}{|S_d|}, \tag{5}$$

where S_d is the set of sentences of d and S_{used} is the set of marked sentences. No LLM is run to produce these labels. When we need true utility for evaluation, we compute Eq. (1) with a source LLM. The no-context answer is generated once per query and reused for all passages of that query.

Optionally, $\hat{u}$ can be calibrated when labels from the target LLM are available for k passages. We fit α and β by linear regression on these k examples and use

$$\hat{u}' = \alpha \cdot \hat{u} + \beta. \tag{6}$$

This changes the scale of $\hat{u}$, which matters for the threshold τ. It does not change the ranking as long as $\alpha > 0$, because the order of the scores stays the same. Calibration is not used in the main results.

The two LLM-based steps, hypothetical answer generation and logit probing, share one model M: Qwen2.5-7B-Instruct loaded in 4-bit NF4 with double quantization and bfloat16 compute. Sharing one copy avoids loading a second 7B model, so the full pipeline fits on a single 8 GB GPU. On Apple Silicon, we use a 4-bit Llama-3.2-3B-Instruct through MLX instead. The NLI model and the regressor are small and run without quantization.

At inference time, MPUP makes one short greedy decode per query (at most 48 tokens) for the hypothetical answer. For each passage it makes one probing pass and one NLI pass, then applies the regressor. The passages are sorted by score, those with a score $\leq \tau$ are dropped ($\tau = 0$ by default), and the top m ($m = 5$ by default) are written as numbered context before the query.

IV. Experimental Setup

A. Datasets

Table III lists the dataset sources that are implemented in the MPUP code base. Only the RAGBench HotpotQA subset is used for the experiments in this paper. The other three sources have working loaders, but they are not evaluated yet.

B. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

C. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.

TABLE III
Dataset sources implemented in MPUP

Dataset	Source (Hugging Face)	Candidate passages per query	Utility label	Status
RAGBench, HotpotQA subset [Friel et al., 2024]	galileo-ai/ragbench, config hotpotqa	Four documents for 388 of the 390 test questions; 100 questions (400 query–passage pairs) are used	Fraction of sentences marked as used (Eq. 5)	Used in experiments
Natural Questions [Kwiatkowski et al., 2019]	google-research-datasets/ natural_questions	Long-answer candidates of the Wikipedia page	Gold short answers, utility from Eq. (1)	Loader only
MS MARCO v2.1 [Bajaj et al., 2016]	microsoft/ms_marco, config v2.1	Ten passages, with the selected ones marked	Gold answers, utility from Eq. (1)	Loader only
TriviaQA, unfiltered [Joshi et al., 2017]	mandarjoshi/trivia_qa, config unfiltered	Wikipedia and web evidence pages	Gold answer aliases, utility from Eq. (1)	Loader only

- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: "Wb/m^2" or "webers per square meter", not "webers/m^2". Spell out units when they appear in text: ". . . a few henries", not ". . . a few H".
- Use a zero before decimal points: "0.25", not ".25". Use "cm^3", not "cc".)

D. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{3}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(3)", not "Eq. (3)" or "equation (3)", except at the beginning of a sentence: "Equation (3) is . . ."

E. LaTeX-Specific Advice

Please use "soft" (e.g., \eqref{Eq}) cross references instead of "hard" references (e.g., (1)). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don't use the {eqnarray} equation environment. Use {align} or {IEEEeqnarray} instead. The {eqnarray} environment leaves unsightly spaces around relation symbols.

Please note that the {subequations} environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you've discovered a new method of counting.

BibTeX does not work by magic. It doesn't get the bibliographic data from thin air but from .bib files. If you use BibTeX to produce a bibliography you must send the .bib files.

LaTeX can't read your mind. If you assign the same label to a subsubsection and a table, you might find that Table I has been cross referenced as Table IV-B3.

LaTeX does not have precognitive abilities. If you put a \label command before the command that updates the counter it's supposed to be using, the label will pick up the last counter to be cross referenced instead. In particular, a \label command should not go before the caption of a figure or a table.

Do not use \nonumber inside the {array} environment. It will not stop equation numbers inside {array} (there won't be any anyway) and it might stop a wanted equation number in the surrounding equation.

F. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word

"alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
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Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more subtopics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

I. Figures and Tables

a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 2", even at the beginning of a sentence.

TABLE IV
Table Type Styles

Table Head	Table Column Head		
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[a]Sample of a Table footnote.

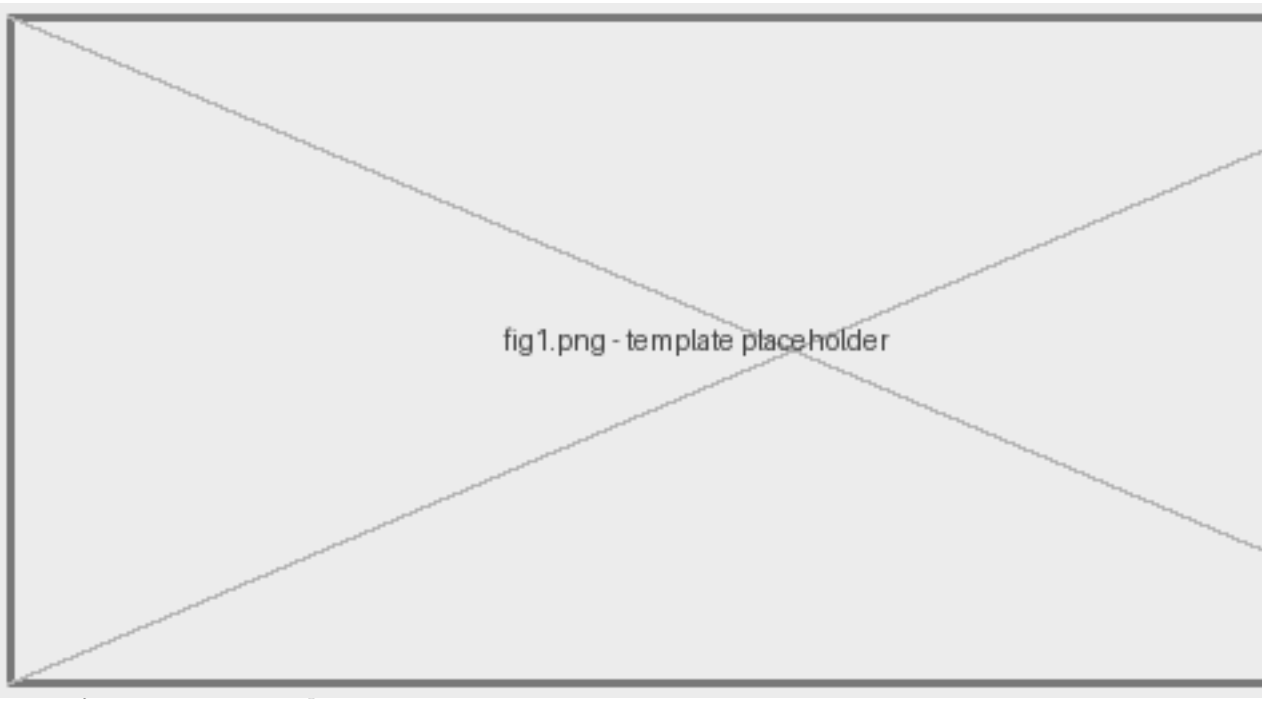


Fig. 2. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try

"R. B. G. thanks. . .". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use "Ref. [3]" or "reference [3]" except at the beginning of a sentence: "Reference [3] was the first . . ."

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors' names; do not use "et al.". Papers that have not been published, even if they have been submitted for publication, should be cited as "unpublished" [4]. Papers that have been accepted for publication should be cited as "in press" [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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